

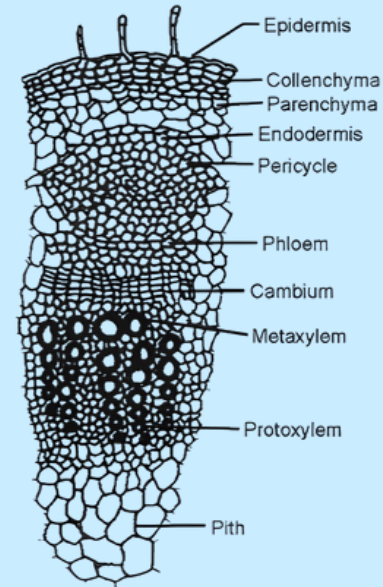
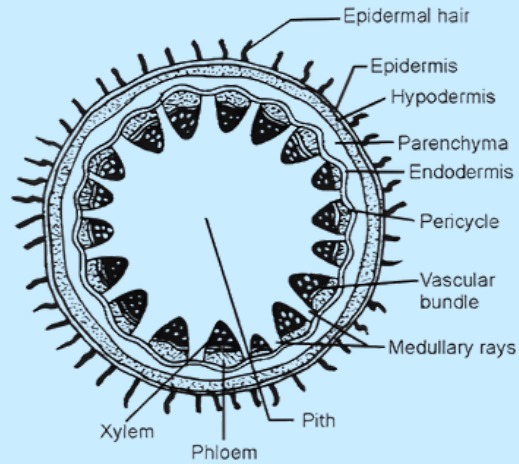
DICOT STEM VS. MONOCOT STEM

Feature	Dicot Stem	Monocot Stem
Epidermis	Single-layered; cuticle thick; may have trichomes	Single-layered; thicker cuticle; usually no trichomes
Hypodermis	Collenchyma present just under epidermis for mechanical support	Sclerenchymatous hypodermis common → stronger mechanical support
Cortex	Parenchyma with chloroplasts; distinct	Often narrow or poorly differentiated
Endodermis	Well developed with starch grains	Absent
Pericycle	Present; Sclerenchymatous semi-lunar patches	Absent

Vascular Bundles	Arranged in a ring, Water-containing cavities absent	Scattered throughout ground tissue, Water-containing cavities present
Vascular Bundle Type	Conjoint Open (cambium present)	Conjoint Closed (no cambium)
Xylem Arrangement	Endarch; protoxylem internal, metaxylem external	Endarch; large vessels in centre, smaller towards periphery
Phloem Arrangement	Sieve tubes + companion cells + phloem parenchyma	Sieve tubes + companion cells; phloem parenchyma usually absent
Medullary Rays	Well developed between vascular bundles	Absent
Pith	Large, well defined	Absent
Bundle Sheath	Rare	Present (sclerenchymatous sheath around bundles)
Ground Tissue	Differentiated as cortex → endodermis → pericycle → pith	Undifferentiated mass of parenchyma

Diagram

DICOT STEM



MONOCOT STEM

